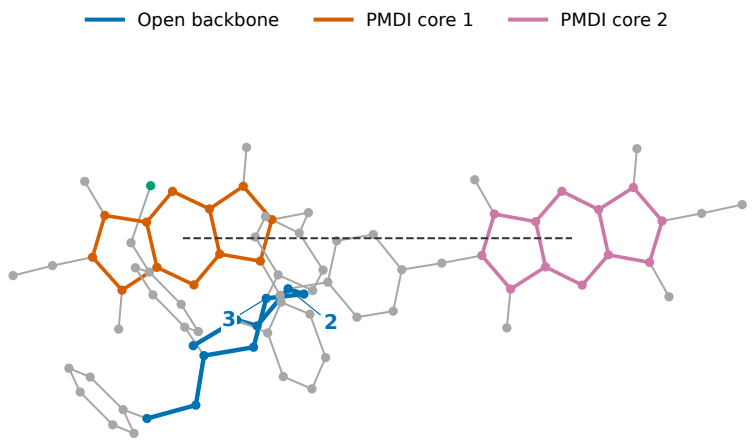
From redox molecules to grafted polymers

Which molecular advantages survive attachment and nearby redox groups?



Actual computed PMDI-grafted open-chain model

Select a useful material, not just a promising molecule.

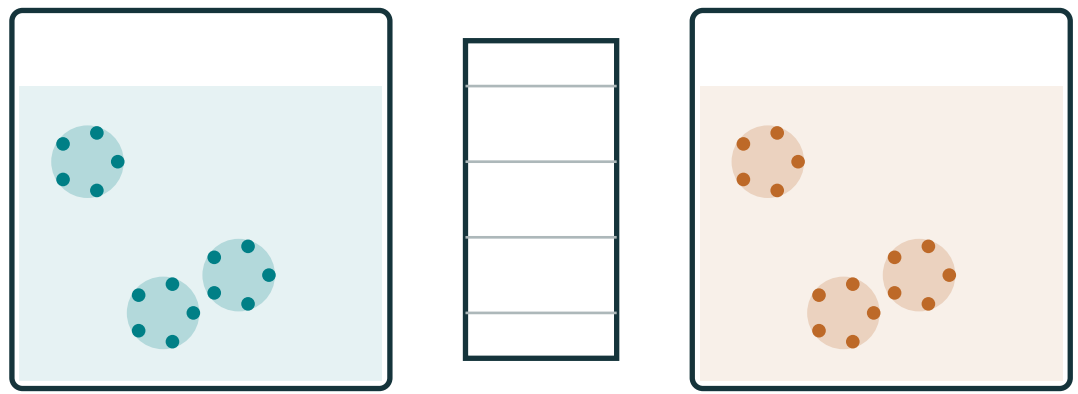
Polymer-bound redox groups offer a route to larger energy carriers

Store charge on redox motifs.

Use a larger carrier to aid separation.

Our route: graft motifs onto an already formed benzyl-chloride polymer.

Conceptual flow-battery architecture



Redox carriers | size-selective separator

The design problem spans both redox chemistry and the polymer environment.

Context: Janoschka et al., Nature 2015, doi:10.1038/nature15746; Montoto et al., JACS 2016, doi:10.1021/jacs.6b06365. Cartoon is illustrative.

A useful motif must work after grafting

01

Useful redox properties

Voltage window
Accessible capacity
Stable charge states

02

Feasible attachment

Exact precursor
Defined reaction conditions
Core retained after grafting

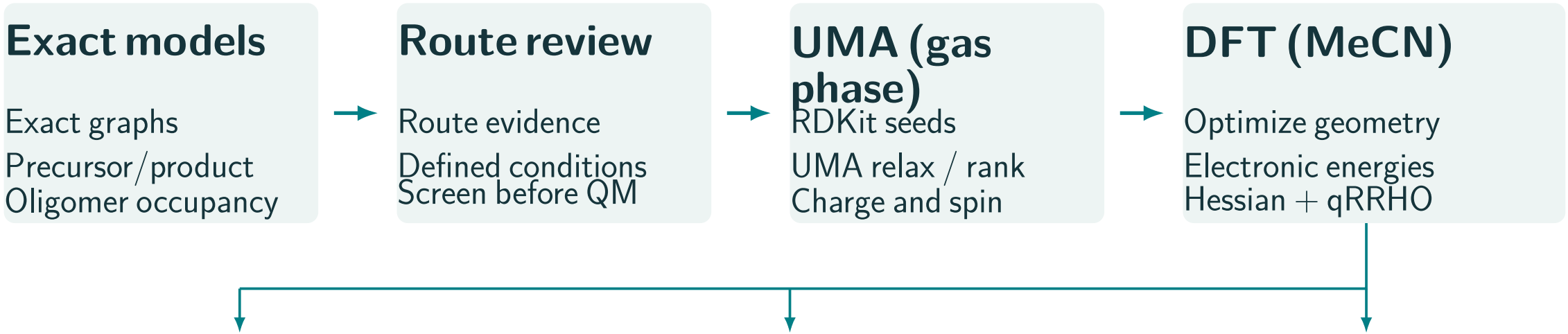
03

Weak unwanted association

Spacing and conformation
Motif contact strength
Intersite coupling

A strong result on one axis cannot compensate for failure on another.

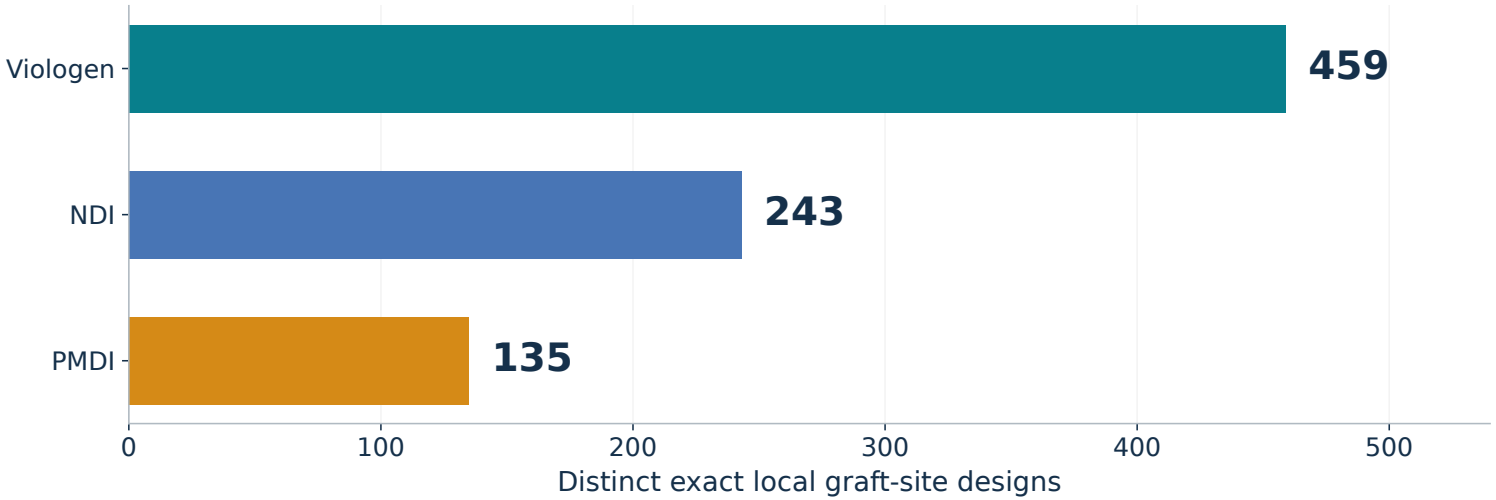
UMA warm starts lead to solution DFT



UMA accelerates the search; balanced DFT free energies support the decision.

UMA is a gas-phase structural accelerator, not the source of the reported solution free energies. Each exact polymer graph is evaluated separately. Nitro numerical qualification remains open; the tested standard-inference profile is separate.

837 exact designs vary chemistry systematically



837

exact local graft models

Design variables

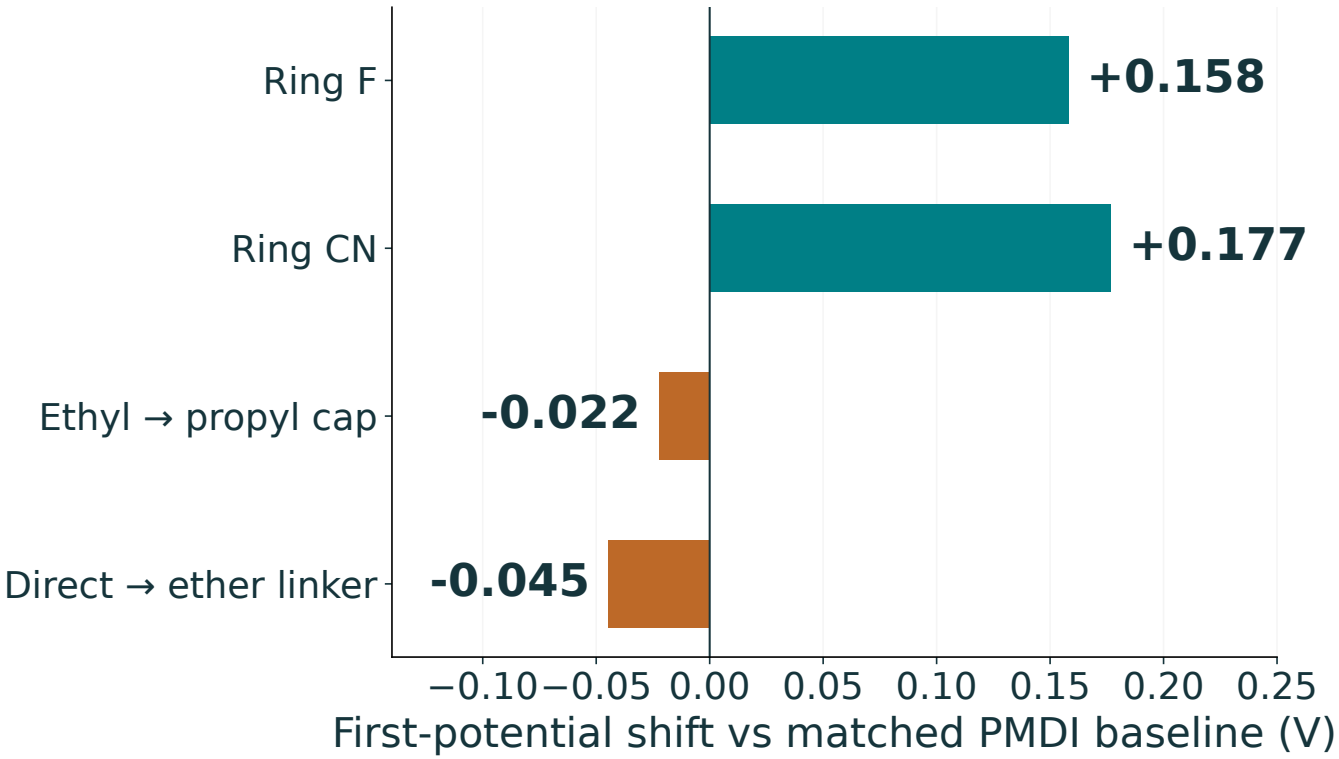
Core
Linker
Cap
Ring substituent

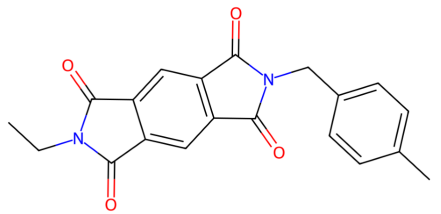
Separately: 75 evaluated candidates; 60 have both computed potentials.

The library is enumerated; it is not an 837-molecule completed quantum screen.

Viologen: 459; PMDI: 135; NDI: 243. Evaluated cohort overlaps 33 library graphs. Comparable BQ/AQ and oxidation-active coverage is still being developed.

PMDI: ring substitution is a stronger voltage lever





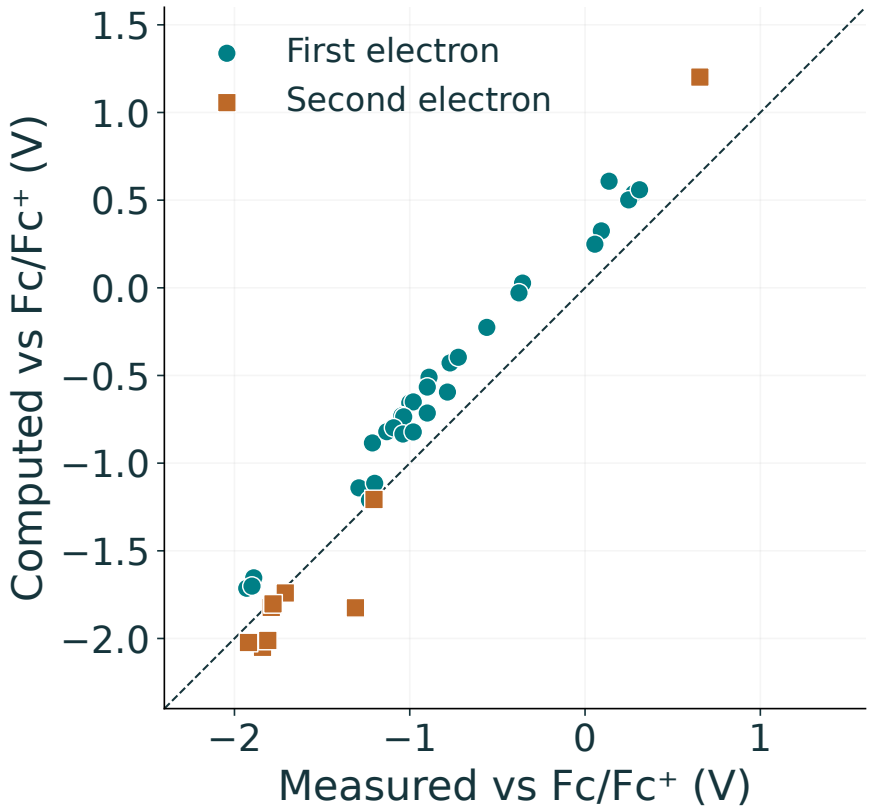
Same baseline

PMDI core
Ethyl cap
Local benzyl graft

Matched comparisons identify tuning levers, not material winners.

Computed first reductions in MeCN, relative to the exact baseline. Four illustrative contrasts from 18 systematic three-core comparisons; all 26 cohort contrasts reconstruct.

The experimental benchmark sets an error scale of about 0.25 V



0.247 V

Unfitted mean absolute error

39 endpoints
30 molecules
7 studies

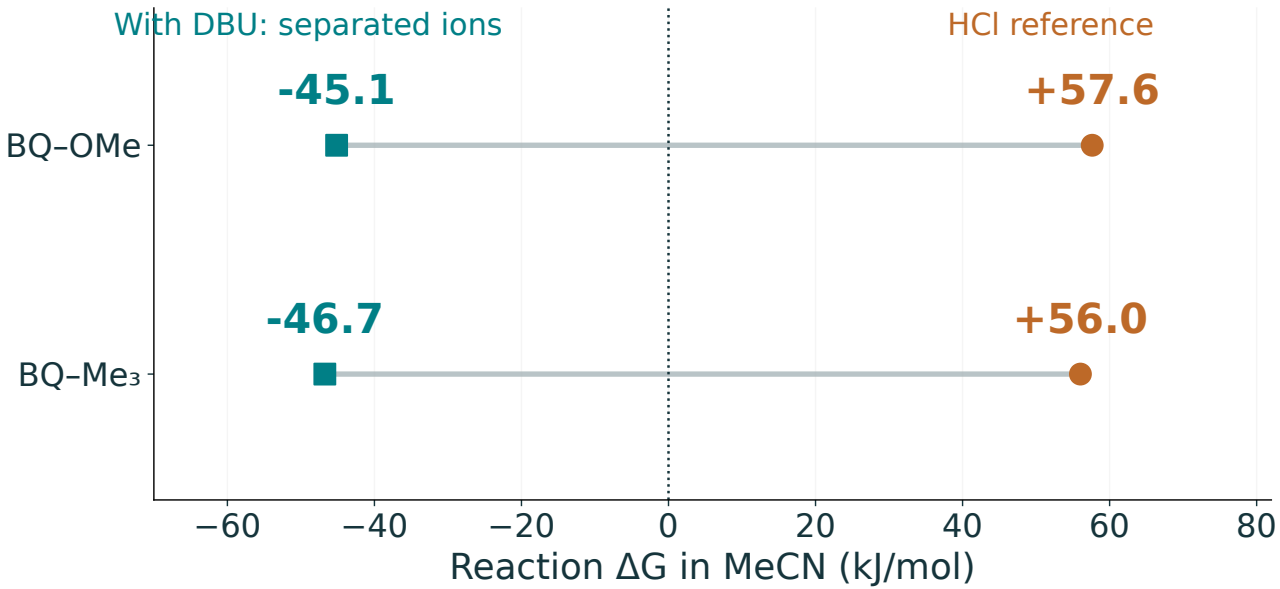
MeCN; RMSE = 0.279 V

Use the benchmark to calibrate confidence in molecular redox predictions.

Retrospective exact-free-molecule panel. Molecule-cluster 95% MAE interval: 0.213–0.286 V. No NDI endpoint; grafted-polymer accuracy is not established.

Grafting ΔG: base choice changes the reaction

$$\text{Motif–OH} + \text{polymer–CH}_2\text{Cl} + \text{B} \longrightarrow \text{polymer–CH}_2\text{–O–motif} + \text{BH}^+ + \text{Cl}^-$$

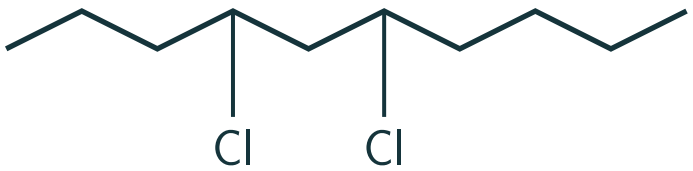


Keep base-required BQs for evaluation; the reaction conditions matter.

Local graft-site calculations, MeCN, 298.15 K. Computed DBU capture is a shared offset of −102.78 kJ/mol; rankings are unchanged. pKa validation exposes an approximately 23 kJ/mol discrepancy. Signs remain conditional; selectivity and yield are untested.

Four matched occupancy states isolate the neighbor effect

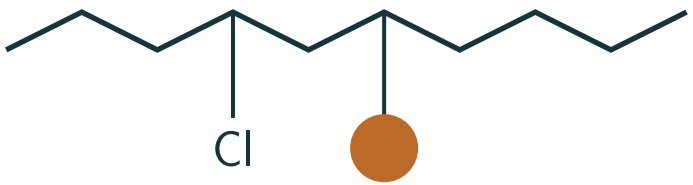
G_{00}



G_{10}



G_{01}



G_{11}

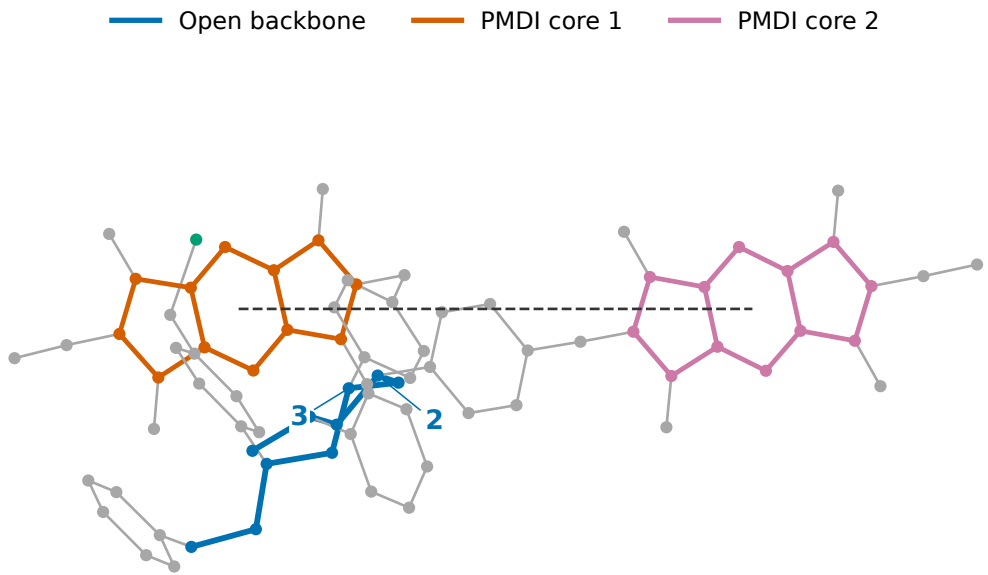


$$\Delta\Delta G = G_{11} + G_{00} - G_{10} - G_{01}$$

Measure how a nearby graft changes the next grafting step.

Schematic of the exact occupancy cycle, not an atomistic drawing. All backbones are open chains. Cl denotes an unreacted benzyl-chloride handle; colored circles denote grafts. Actual loading and spacing are unknown.

PMDI: the sampled neighbor effect is modest



3D core-centroid distance: 16.30 Å; plane angle: 69.0°

−1.86

kJ/mol
Neighbor $\Delta\Delta G$

16.30 Å

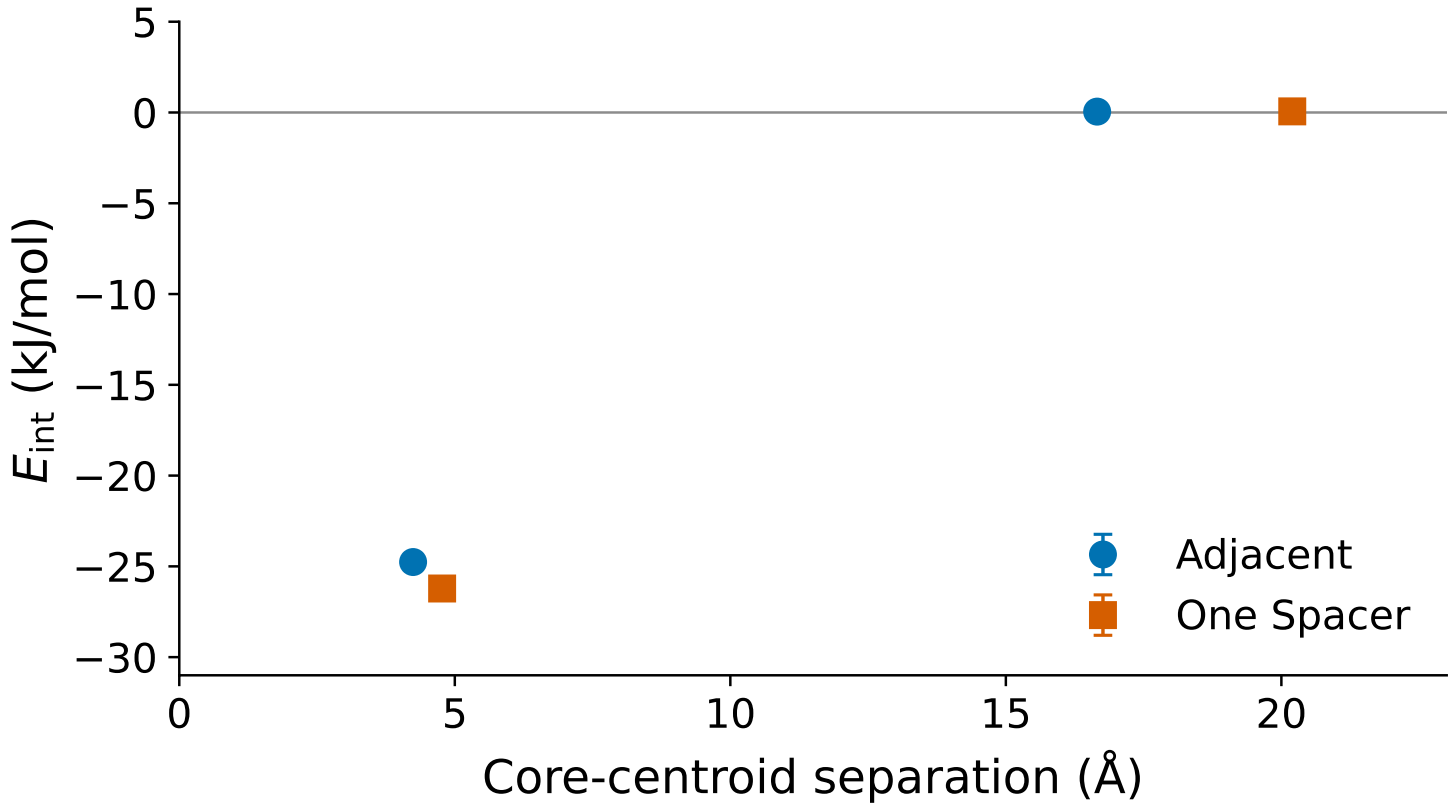
Core separation

Four matched free energies; one minimum per graph.

The finite PMDI cycle quantifies a polymer-context effect.

Exact ethyl-PMDI design. Local redox: $E_1 = -1.076$, $E_2 = -1.977$ V vs Fc. First-site attachment is 5.34/6.11 kJ/mol less favorable than the capped model. Whole-chain nonadditivity includes backbone and solvent response.

A spacer can still permit a strongly attractive close contact



−25 to −26

kJ/mol
Close phthalimide poses

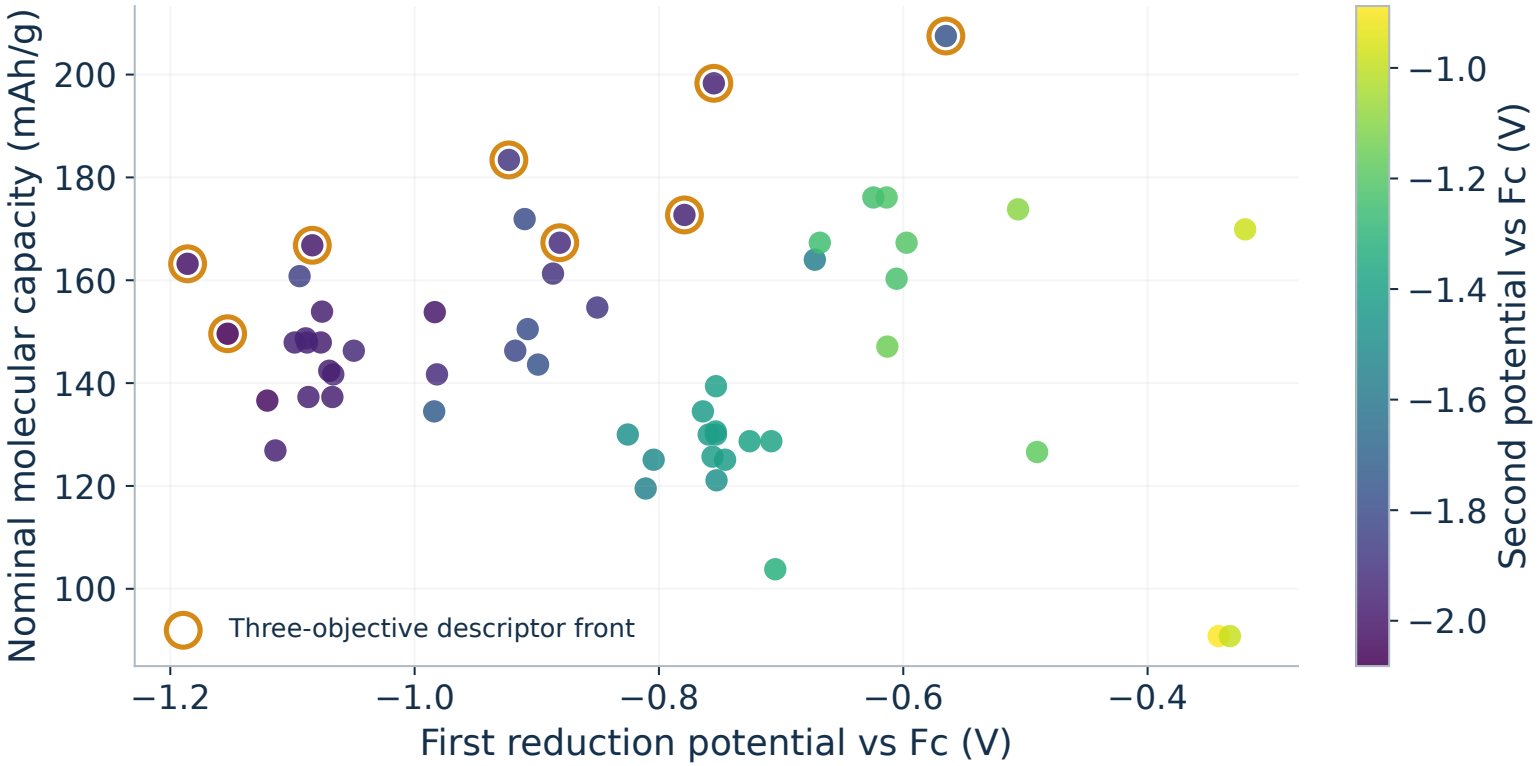
≈ 0

kJ/mol
Separated poses

Design for weak unwanted interaction; sequence spacing alone is not a reliable proxy.

Frozen capped-fragment electronic interactions in the MeCN model, $E_{AB} - E_A - E_B$. Phthalimide is a control, distinct from PMDI. No association free energy, conformer population, basis-superposition correction or electronic coupling is established.

A Pareto leader must still pass every material requirement



8

Molecular descriptor leaders

0

Demonstrated all-axis candidates

Use the descriptor front to prioritize questions, not to claim a synthesis shortlist.

Projection of the three-objective comparison over 56 reduction-active local models. Nominal molecular capacity excludes polymer/loading penalties. Route, stability, accessibility, occupancy and interaction evidence remain incomplete.

Three design lessons now guide the next candidates

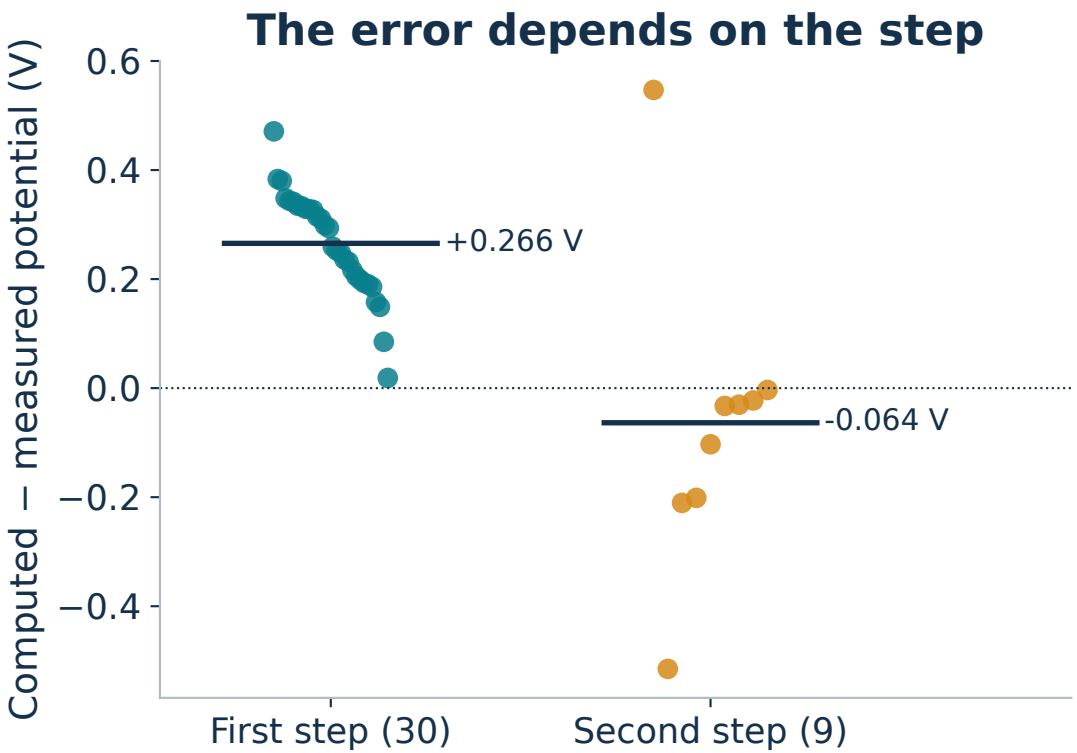
1 **Tune redox chemistry systematically.**
Ring substitutions shift voltage; benchmark error is about 0.25 V.

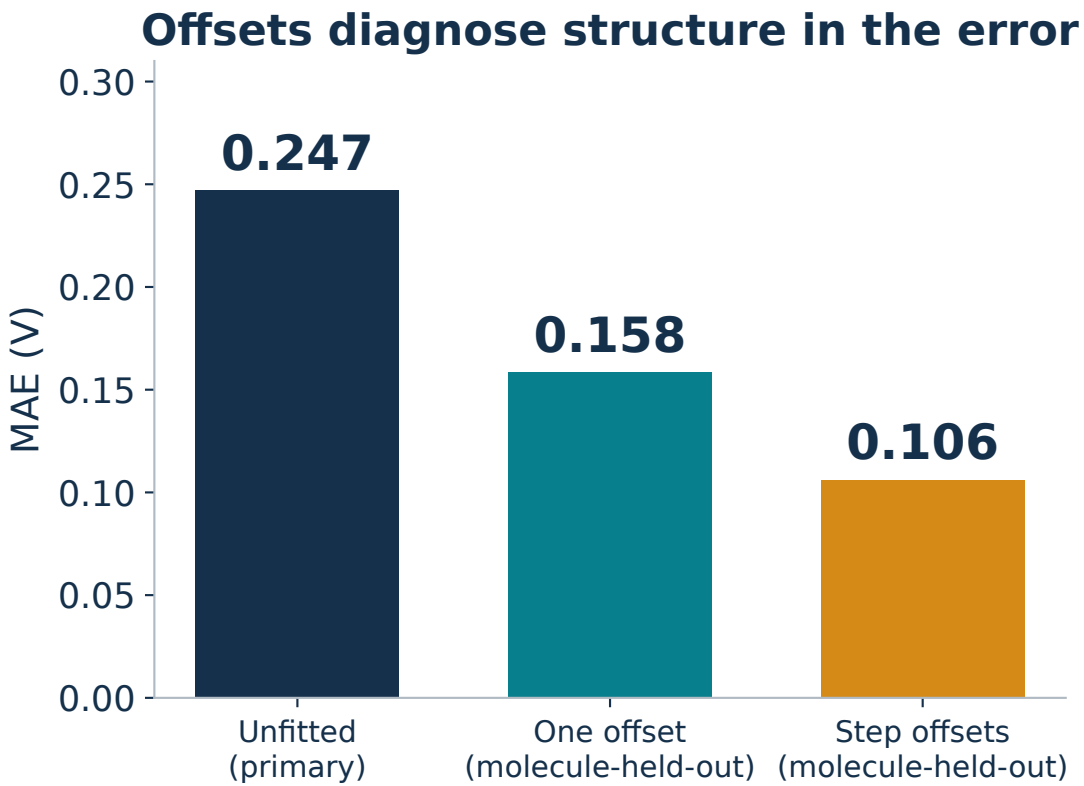
2 **Evaluate a defined attachment reaction.**
Base and speciation affect feasibility; a reference reaction is insufficient.

3 **Test the local polymer environment.**
A spacer does not guarantee weak interaction.

Next: broader chemistry and complete polymer-context evidence.

Benchmark bias and numerical sensitivity





Offsets are internal diagnostics; primary benchmark statistics remain unfitted.

Molecule-held-out MAE: 0.158 V, one offset; 0.106 V, separate step offsets. Five neutral SVPD controls move potentials by −11.7 to −35.1 mV. Neither diagnoses all remaining error sources.

What is computed, and what has external validation?

Quantity	Evidence	Limit
Redox potential	39 exact-parent MeCN endpoints	No polymer or NDI accuracy claim
Grafting ΔG	Balanced reactions; experimental base-pKa comparisons	Speciation, selectivity, barriers
Polymer neighbor effect	Complete finite PMDI cycle	Broader ensembles pending
Local motif interaction	12 triplets / 36 electronic states	No binding G or coupling

Numerical consistency, experimental accuracy and sampling robustness answer different questions.

Conformer sampling is being tested explicitly

Declared expansion

20 starts for each of
12 graph/configuration sets.

Two backbone configurations.
Occupancy and spacing varied.

First completed diagnostic

One empty-backbone configuration:
10 → 20 checked conformers.

Minimum E and G unchanged;
ensemble G changes by −1.03 kJ/mol.

One component meets the declared 2 kJ/mol batch criterion; full occupied cycles remain open.

The new analysis uses a separately identified smooth-surface MeCN model; it cannot be mixed with the primary PMDI cycle. Global sampling convergence is not established.

Reproducibility and model qualification remain explicit

Available

Code and exact chemical identities

Checkpointed calculations

Frozen evidence and raw-state packages
Relocated arithmetic/Hessian replay

Remaining

Public data DOI and metadata

Broader deployment portability

Nitro force-repeatability qualification
Complete material-level assessment

UMA is an upstream accelerator; reported solution free energies retain the checked DFT protocol.

Current H100 Nitro repeatability controls fail the stated force threshold; installation is not numerical qualification. V100 uses a separately tested standard-inference fallback.